

ChronoPDE

A controlled study of boundary-aware continuous-time neural operators

A reproducible PyTorch investigation of FFT and DCT neural operators for parameterized two-species reaction-diffusion dynamics.

Outcome: the main Week 6 gate did not pass. Objective alignment improved both models sharply and DCT was lower-error on all 16 matched diagnostic samples, but neither reached the predeclared 0.01 velocity-nRMSE threshold.

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Controlled Study & Failure Analysis

1. Problem and contribution

ChronoPDE asks whether a cosine spectral backbone aligned with homogeneous Neumann boundaries improves a continuous-time learned velocity field relative to a parameter-matched Fourier control. Both vector fields are conditioned on physical parameters and time, and both are integrated with RK4.

Implemented system

The repository contains an independently implemented reaction-diffusion simulator, a deterministic 720-trajectory 64 x 64 dataset, temporal masks, orthonormal DCT transforms, quintic spline targets, fixed-step integrators, autoregressive U-Net and FFT-FNO baselines, and approximately 1.95M-parameter continuous-time FFT and DCT models.

Original hypotheses

H1: continuous-time velocity learning improves behavior with sparse or irregular observations. H2: a DCT backbone improves boundary and spectral behavior under Neumann conditions. The execution plan required a four-trajectory gate before sparse-time or OOD testing.

Recovery contribution

After the gate failed, the project followed its stop rule. It audited targets, tested matched optimizer and loss controls, corrected an unrepresentative batch, and aligned the objective with the gate metric. The resulting evidence separates dataset validity, objective mismatch, architecture behavior, and unsupported claims.

Component	Frozen choice
Grid and state	64 x 64; two channels; no-flow boundary
Models	CT-FFT 1,973,657; DCT 1,951,125 parameters
Diagnostic batch	4 trajectories x intervals 0, 33, 66, 99
Optimization	AdamW; lr 3e-4; zero decay; 5,000 steps
Gate	>=1,000x loss reduction and median nRMSE <=0.01

2. Protocol and claim discipline

The diagnostic protocol was fixed before inspection: seed 0; the same 16 sample identities for both backbones; gamma 0; batch size 16; constant learning rate $3e-4$; no weight decay; evaluation every 100 steps; and a maximum of 5,000 steps.

Metric contract

The aligned objective is the mean across samples of squared full-field error divided by target energy. The gate uses per-sample full-field velocity nRMSE. Historical code used `torch.median`, which returns the lower middle observation for 16 values. This report preserves that decision and also states the conventional interpolated median; neither definition changes the outcome.

Evidence chain

Seven archives are identified by SHA-256 and source commit. Only lightweight JSON, CSV, configuration, and image evidence is committed. Checkpoints, datasets, nested repositories, and ZIP files remain outside version control.

Permitted claims

The target audit passed; alignment improved both models; and DCT was lower-error on every matched sample in this diagnostic. The analysis does not claim that DCT generalizes better, that ChronoPDE passed Week 6, or that sparse-time and OOD performance was validated.

Stop rule

Because both models missed the registered threshold after the complete budget, the selected route is `stop_and_document_model_or_conditioning_limitation`. No 10,000-step comparison, architecture variant, production retraining, or OOD run is treated as authorized by this experiment.

3. Diagnostic sequence

The initial resampled smoke run reduced loss without reaching the desired velocity or rollout behavior. A CPU target audit then tested whether the spline derivative, physical time scaling, spectral truncation, or normalization contract was corrupt.

Target audit

All 400 fixed samples were finite. Median spline-to-PDE velocity nRMSE was 0.00151, far below the 0.15 repair threshold. Twelve DCT modes retained about 79.2% of target energy, while 20 modes retained about 97.4%.

Matched mechanics and representative sampling

Shared learning-rate and loss controls left FFT and DCT closely matched and below the required absolute accuracy. The first batch was then found to contain only the first trajectory. A balanced follow-up selected intervals 0, 33, 66, and 99 from each of four trajectories. Both backbones again failed despite large global-loss reductions: best velocity nRMSE was 0.1205 for FFT and 0.2480 for DCT.

Why align the objective?

Global MSE gives high-energy examples greater influence, while the gate weights each sample through its own target energy. The production auxiliary spectral term also measures only the lowest 12 x 12 modes. The final diagnostic changed only its objective to full-field relative error, leaving production defaults untouched.

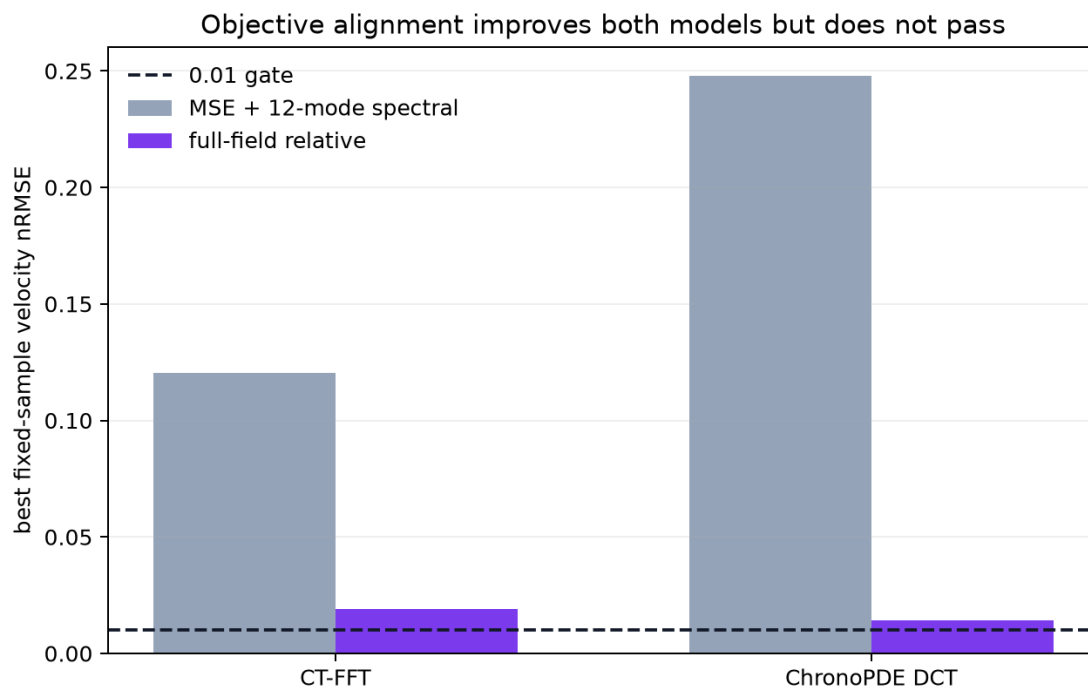


Figure 1. Objective alignment improved both models sharply, but both bars remain above the unchanged 0.01 gate.

4. Loss-alignment result

Model	Best step	Loss reduction	Gate nRMSE	Decision
CT-FFT	4100	188684x	0.01895	Fail
ChronoPDE DCT	5000	7980x	0.01421	Fail

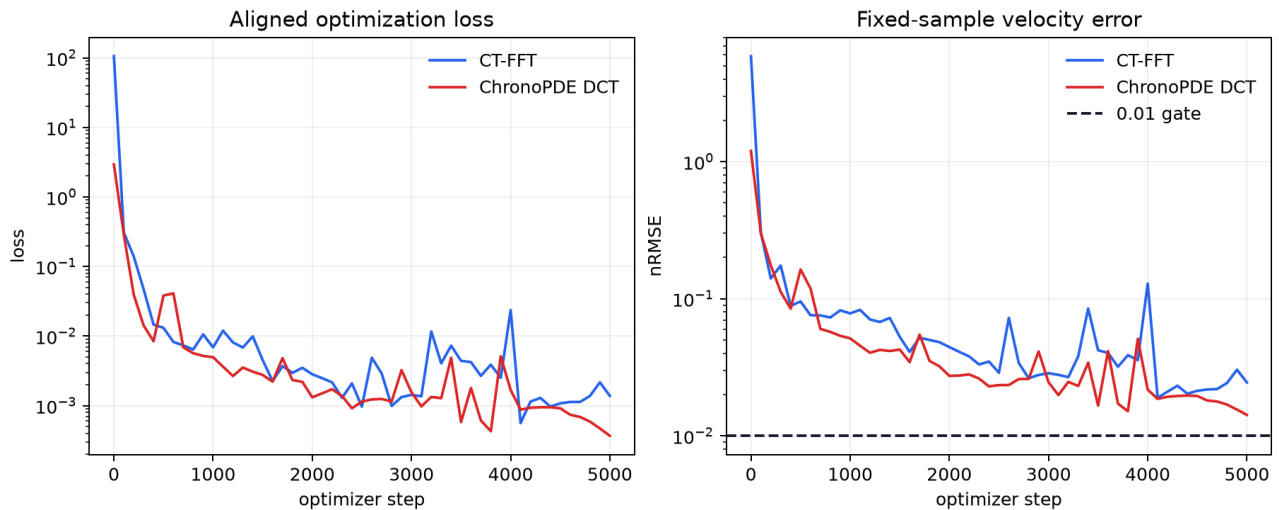


Figure 2. Both models exceed the loss-reduction requirement, but their selected velocity errors remain above the gate. FFT's best eligible step is 4100; DCT's is 5000.

The conventional medians are 0.01950 for FFT and 0.01514 for DCT. Reporting that distinction improves metric transparency but does not convert either result into a pass.

5. Matched sample analysis

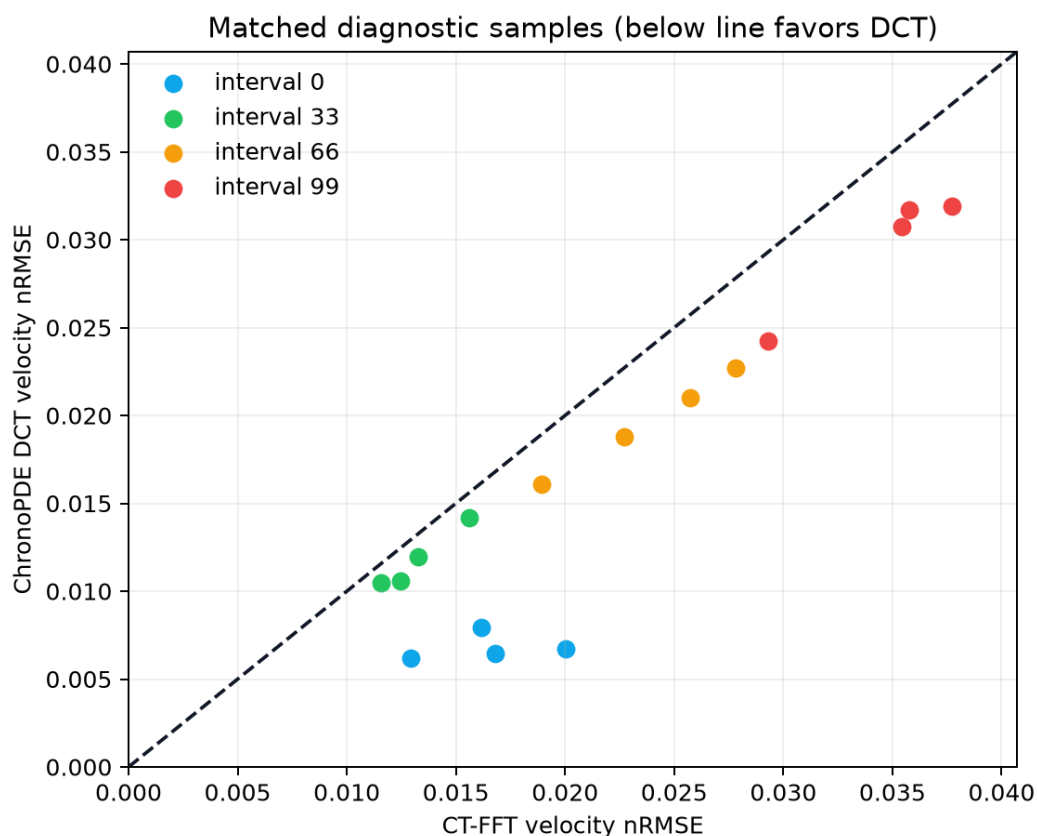


Figure 3. Every marker lies below the equality line: DCT has lower velocity nRMSE for all 16 matched trajectory-interval identities at the selected checkpoints.

The paired mean DCT-minus-FFT difference is -0.00503. A deterministic 10,000-resample paired bootstrap gives a 95% interval of [-0.00668, -0.00354]. The interval describes this selected diagnostic batch; it is not a population-level or multi-seed generalization result.

Channel behavior

At the selected DCT step, the historical channel medians were 0.01837 for u and 0.00864 for v . FFT recorded 0.02938 and 0.01450. Late intervals remained among the hardest examples, motivating future work on conditioning and target balancing.

6. Context, limitations, and conclusion

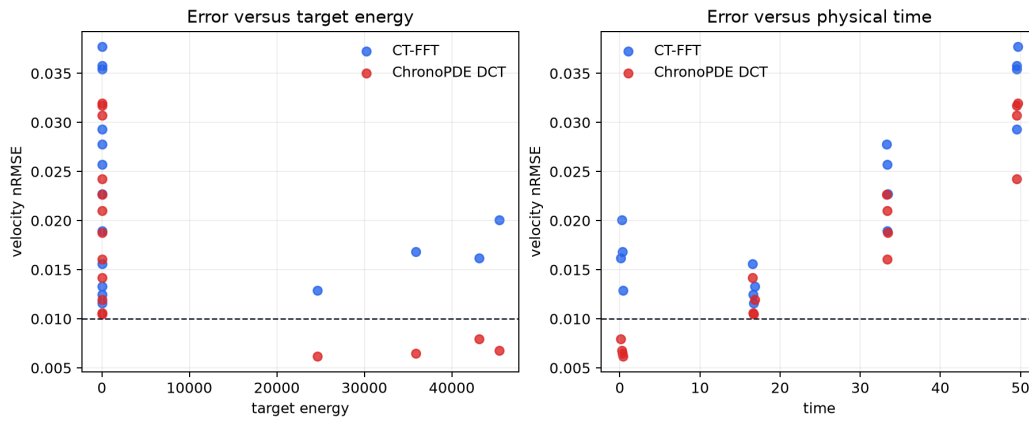


Figure 4. Diagnostic error varies with physical time and target energy.

Exploratory ID context

The 100-trajectory exploratory evaluations were stable and beat persistence. CT-FFT recorded final nRMSE 0.4219; DCT recorded 0.4668. U-Net AR and FFT-FNO AR recorded 0.0419 and 0.0789. These values are context only: the continuous-time runs used unequal histories, and the original AR training summaries also recorded unmet gates.

Conclusion

The project does not establish the original headline claim. It does establish a reproducible failure-analysis result: the target construction is numerically consistent, global objectives can hide large per-sample relative errors, and the boundary-aware DCT model responds favorably to an aligned loss without clearing a strict memorization threshold. This is a scientifically meaningful outcome because the stopping rule, failed threshold, and unsupported experiments remain visible.

References

- [1] Hou, Huang, and Perdikaris. CFO: Learning Continuous-Time PDE Dynamics via Flow-Matched Neural Operators. ICLR 2026.
- [2] Takamoto et al. PDEBench: An Extensive Benchmark for Scientific Machine Learning. NeurIPS Datasets and Benchmarks, 2022.
- [3] Li et al. Fourier Neural Operator for Parametric Partial Differential Equations. ICLR 2021.